

LLM Odyssey: A Game-Based Platform for Teaching LLM Engineering Concepts

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CS graduates get little hands-on LLM engineering practice.

Practitioners name four competencies that programs do not consistently develop.

Prompt engineering

Retrieval-augmented generation

Fine-tuning

Production deployment

Many existing tools focus on individual concepts.

LLM Odyssey integrates them into a scaffolded learning pathway.

What exists

OpenAI Playground

Hugging Face Spaces

Lectures and textbook examples

What Odyssey adds

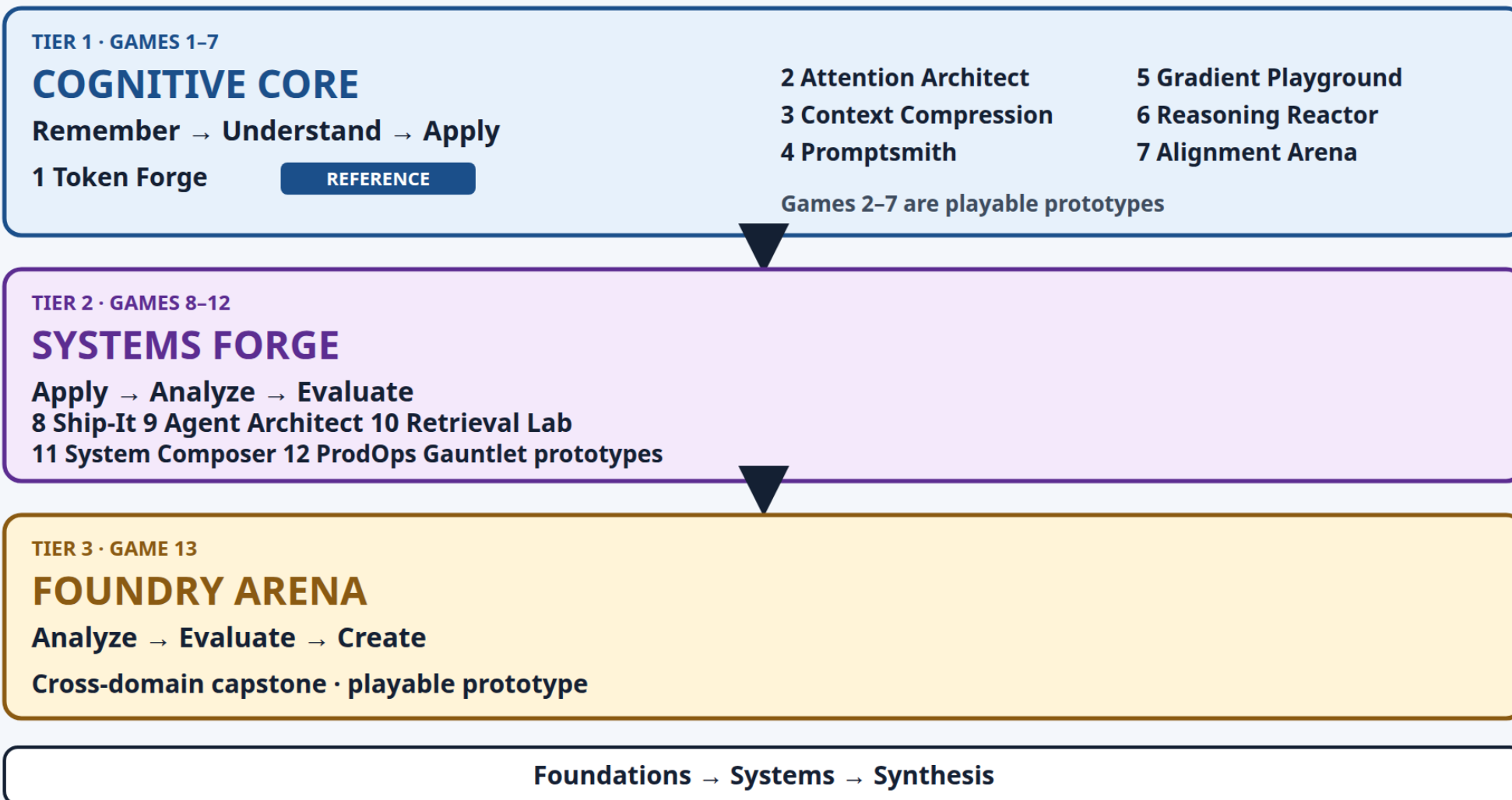
Progressive difficulty

Pedagogical scaffolding

A path from tokenization to a design brief

13 games move from foundations to systems to synthesis

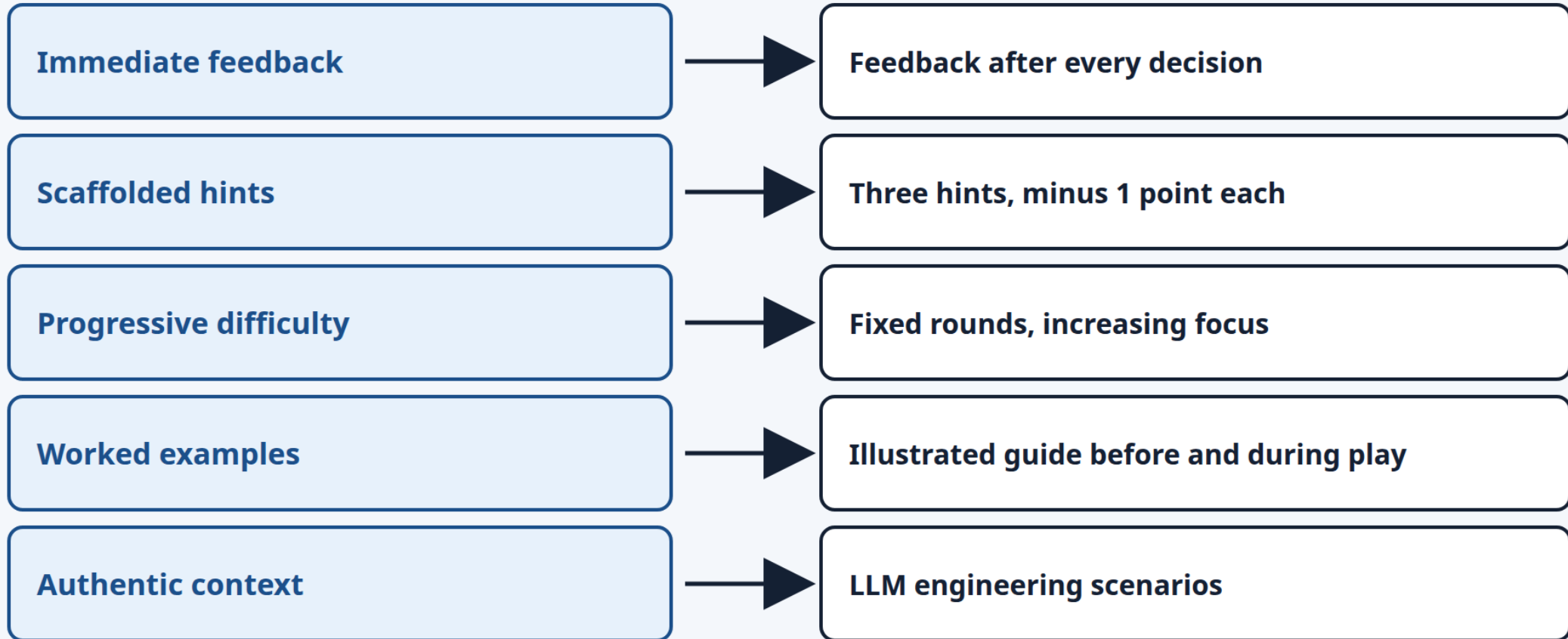
Three learning tiers



Later tiers open at a 70% instructional mastery benchmark. That is not a completion threshold and not professional expertise.

Learning-science principles become game mechanics

Learning science → game mechanic



Black & Wiliam · Vygotsky · Sweller · Bloom. Practice support, not a measured gain.

Students make an engineering decision — and immediately see why it mattered

```
_def | _refund_total | ( | items | : | _list | [str] | ) | _→ | _int:
```

Tokens
10

Vocabulary efficiency
86%

Cost for 250,000 calls
\$37.50

_def _refund_total (items : _list [str]) _→ _int:

1 Scenario pieces

2 Count and cost

Preview only. Locking the round runs the scorer. This round is locked.

Hints

Hints used 0/3. Each hint subtracts 1 point from this round after scoring, to a floor of 0.

Feedback

Your decision. SentencePiece (10 authored tokens)

Result. Correct

Why. Keeping the identifier intact reduced the authored token count for this signature. Code is a different distribution from prose. This comparison was written so SentencePiece keeps the identifier intact. That is a teaching choice, not a benchmark of the SentencePiece library.

What mattered in this scenario? A split identifier is harder to align with the code the caller meant, and it is longer.

Why the alternatives were weaker. Rows that slice the name into many pieces can look thorough and still fail the goal of keeping the identifier intact.

Trade-off. Splitting snake_case and punctuation makes the prompt longer and harder to align with the code. A compact row can keep the name intact.

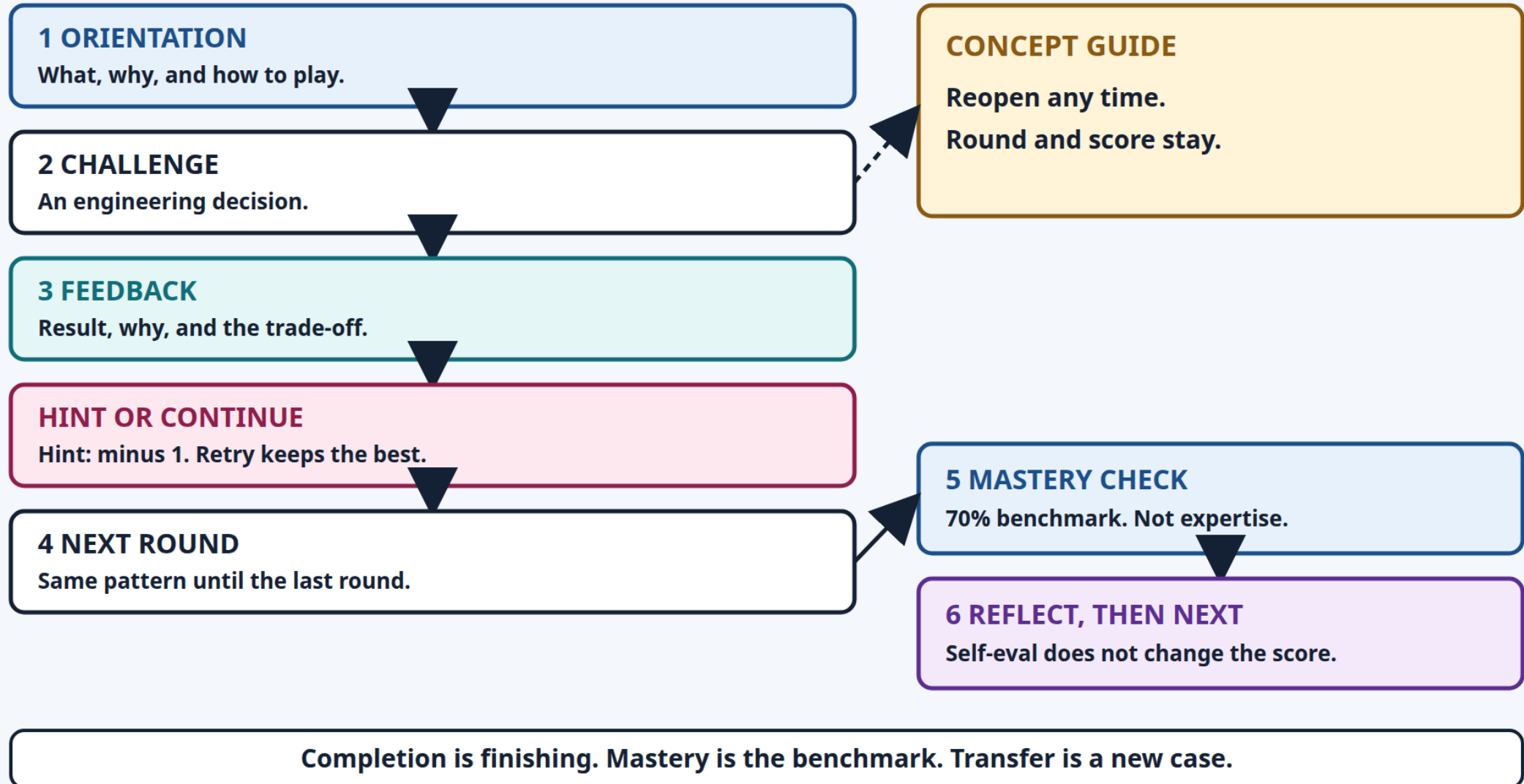
Engineering takeaway. Code is a different distribution from prose, so the preferred segmentation can change.

3 Learner choice

4 Immediate feedback

Each game follows the same scaffolded learning cycle

How one game teaches



Educators can inspect the instructional design before assigning a game

Learning objectives				Current game status		Bloom + estimated time		
#	Game	Tier	Students learn	Status	Minutes	Bloom	Mastery	
1	Token Forge	Cognitive Core	See how text becomes the discrete units processed by a language model. Compare subword segmentation strategies and reason about vocabulary, sequence length, multilingual text, code, context use, and token-related cost.	Reference	20	remember, understand, apply	70%	
2	Attention Architect	Cognitive Core	Compute a tiny self-attention pattern and read which token receives the weight.	Prototype	25	understand, apply	70%	
3	Context Compression	Cognitive Core	Fit source text into a token budget without dropping the fact the task needs.	Prototype	20	understand, apply	70%	
4	Promptsmith	Cognitive Core	Match a prompting strategy to a task and read the quality, token, and failure tradeoff.	Prototype	25	apply, analyze	70%	
5	Gradient Playground	Cognitive Core	Tune learning rate, epochs, batch size, and full fine-tuning versus LoRA on an educational loss model.	Prototype	25	understand, apply, analyze	70%	
6	Reasoning Reactor	Cognitive Core	Assemble a visible reasoning workflow: decomposition, checks, samples, and temperature.	Prototype	25	apply, analyze	70%	
7	Alignment Arena	Cognitive Core	Rank replies with an explicit rubric for helpfulness, safety, and factuality.	Prototype	25	understand, analyze, evaluate	70%	

The next question is empirical: does Odyssey improve learning?

Evaluation flow · documented, not proven

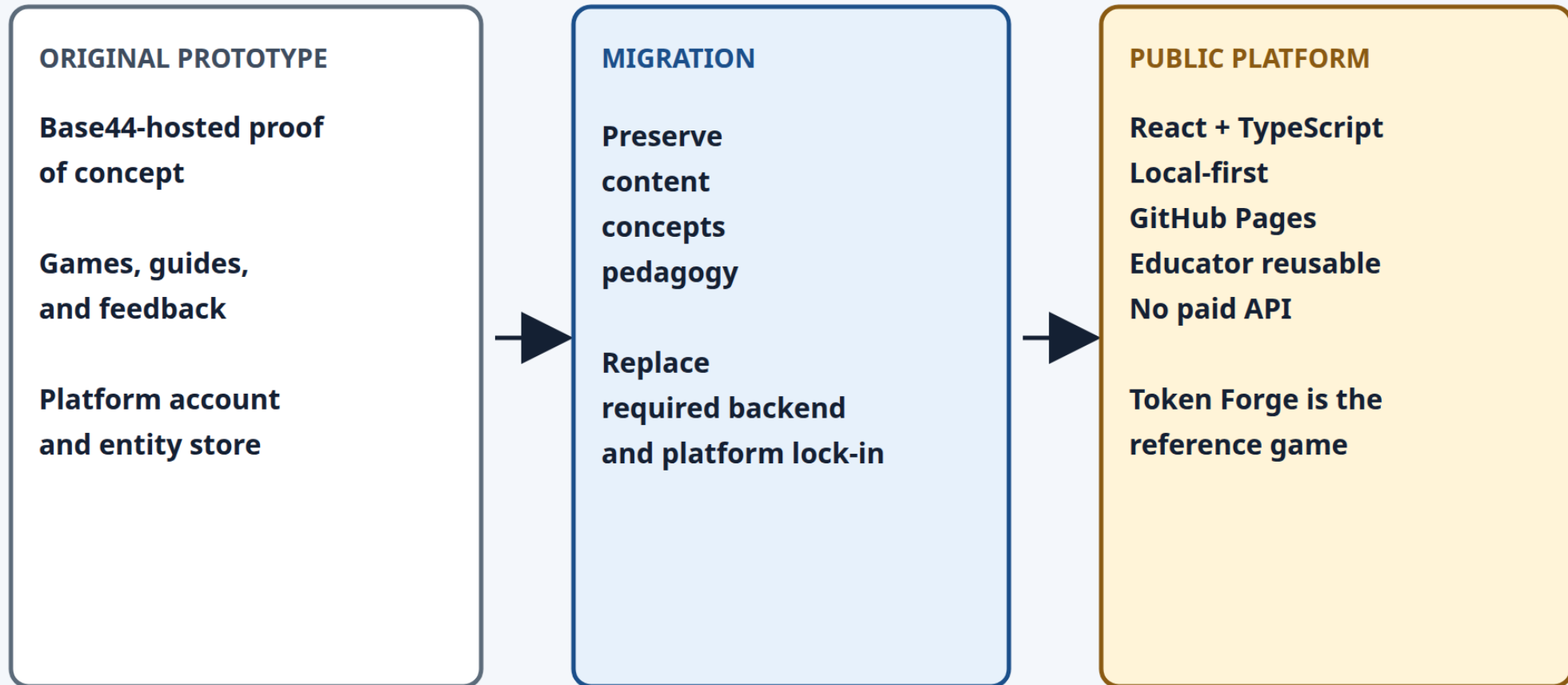


Questions for a future study: knowledge, engagement, feature use, prior expertise.

Learning effectiveness has not yet been established.

From working prototype to reusable open-source teaching platform

From prototype to public platform



Product evolution. The prototype supplied the curriculum. This release stands alone.

<https://piaattufts.github.io/llmodysey/>

<https://github.com/piaattufts/llmodysey>

Backup • Earlier deployment performance targets

< 16 ms

interaction response

< 100 ms

page load

60 fps

animations

0

availability incidents

Game logic runs in the browser, so play does not wait on a server round trip.

From an earlier deployment report. Not a new benchmark of this public release, and not a learning result.

Backup • Current open-source app

LLM Odyssey

13 games · 3 learning tiers · Browser-based · No paid API required · Open source · Educator reusable

Learn

AI Games Arcade

Concept Index

Progress

Apply

The Foundry

Assess

Pre-assessment

Post-assessment

Feedback

Teach

Educator Guide

Demo

Demo Mode

Your journey

0 of 13 games at the mastery threshold

Current game: Token Forge

Open the progress record

Game list

Cognitive Core

1. Token Forge · Reference

2. Attention Architect · Prototype

3. Context Compression · Prototype

The Odyssey · Master the Machine

LLM Odyssey

Learn LLM Engineering Through Interactive Games

LLM Odyssey is an open-source, browser-based learning environment that turns core LLM engineering concepts into interactive decision-making activities.

Students move from foundational topics such as tokenization and attention to retrieval, agents, production operations, and system design.

13 interactive games

3 learning tiers

No paid API required

Open source

Local-first

Educator reusable

Token Forge is the reference implementation. The other twelve games are playable prototypes. Their scores are calculated, and each card shows that status. Nothing here calls a paid model API.

Start Learning

Explore as Educator

Take the Demo Tour

Continue: Token Forge

View on GitHub

0 of 13 games mastered. Recommended next: Token Forge.

Optional pre-assessment

Practice ahead. Opens later tiers before the mastery counts are met. Listed prerequisites stay visible either way.

How the Odyssey works

1. Understand

Read why the game exists, what you will do, and what the screen is actually calculating.

2. Experiment

Make a decision on a scenario. The examples stay the same for every learner.

3. Get feedback

See why the choice fit, why the alternatives were weaker, and what a real system would require.

4. Reflect

Answer a short prompt. The reflection does not change the score.

5. Apply

Use the concept guide and, later, a design brief in Foundry Arena.

6. Progress

Completion, the score, and the mastery threshold are recorded separately. Transfer is whether you can explain a new case.

threshold

Current game: Token Forge

Open the progress record

Game list

Cognitive Core

1. Token Forge · Reference

2. Attention Architect · Prototype

3. Context Compression · Prototype

4. Promptsmith · Prototype

5. Gradient Playground · Prototype

6. Reasoning Reactor · Prototype

7. Alignment Arena · Prototype

Systems Forge

8. Ship-It Simulator · Prototype

9. Agent Architect · Prototype

10. Retrieval Lab · Prototype

11. System Composer · Prototype

12. ProdOps Gauntlet · Prototype

Foundry Arena

13. Foundry Arena · Prototype

How this round is scored

Minimize tokens while keeping the function name recognizable.

def refund_total(items: list[str]) → int:

BPE

SELECT

Pairs have not fully merged the snake-case name in this row.

WordPiece

SELECT

Continuation marks appear inside the name and the brackets.

SentencePiece

SELECTED

The identifier and the annotation stay in fewer authored pieces.

Unigram

SELECT

Closer than BPE, but the arrow and brackets are still separate pieces.

_def | _refund_total | (| items | : | _list | [str]) | _→ | _int:

Tokens

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Vocabulary efficiency

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_def

_refund_total

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items

:

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)

_→

_int:

Preview only. Locking the round runs the scorer. You can still change the selection.

Hints

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Reveal next hint

Lock in this round

LLM Odyssey

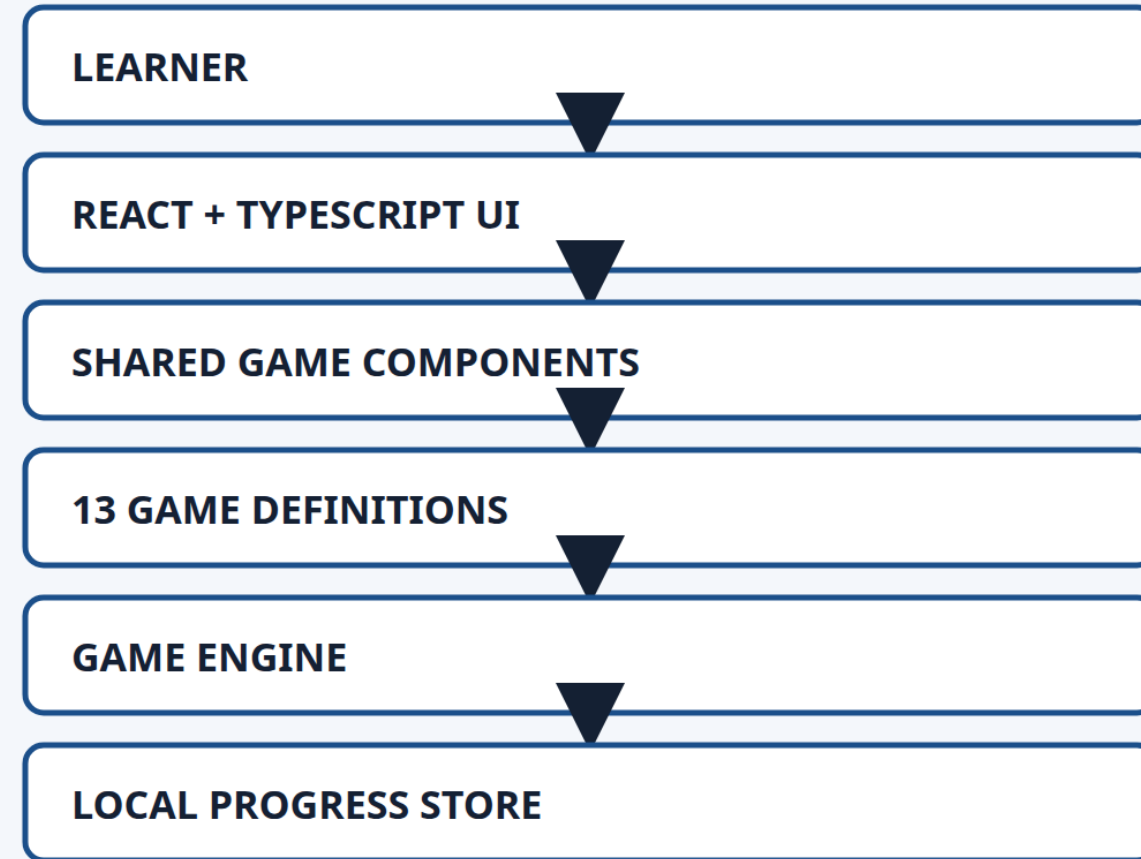
Homepage

Token Forge, reference game

13 games • 3 tiers • no paid API required for core gameplay

Backup • Core gameplay needs no backend and no paid model API

Standalone architecture



IndexedDB, with localStorage if IndexedDB is unavailable.

OPTIONAL

Off unless configured

Analytics adapter

Supabase only with research mode and a project URL

LLM provider adapter

Default is a mock.
The 13 games do not call it.

A missing backend does not stop the games.

Core gameplay needs no backend and no paid model API.

Backup • Faculty feasibility feedback, not a learning result

Two faculty reviewers, Winter 2026.

The author was then at Durham College. That is not the current affiliation.

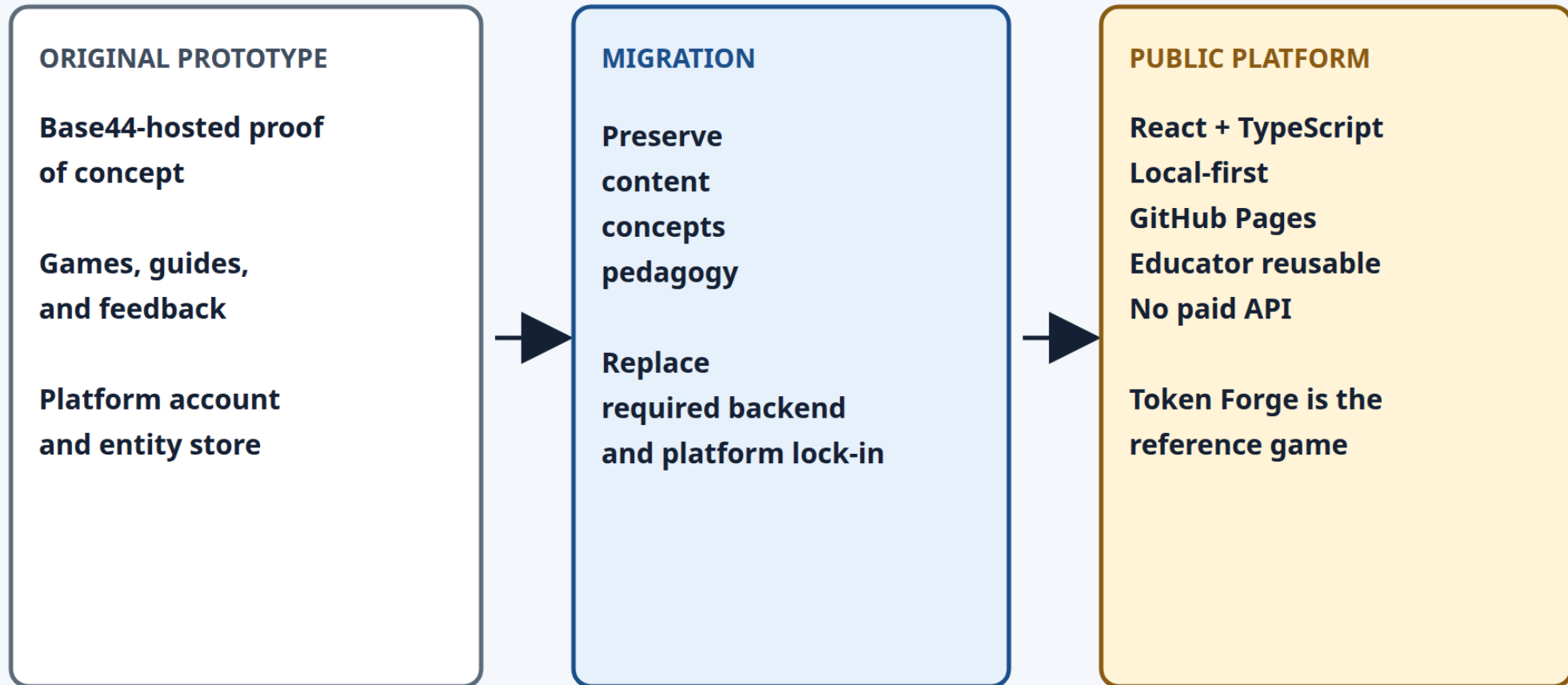
They confirmed the platform runs.

They named adaptive difficulty as the next priority.

This is face-validity feedback. It is not evidence of learning effectiveness.

Backup • Product history, without unlabeled prototype screenshots

From prototype to public platform



Product evolution. The prototype supplied the curriculum. This release stands alone.

Original Base44 screenshots are not in this package. The current app is the public platform.